Research Statement

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My research combines microeconomic theory and experiments to study how people learn from data, how context shapes beliefs and preferences, and how those beliefs influence strategic choices. Across these questions, I treat theory and evidence as close complements: I begin by developing a theoretical framework, then design experiments or use existing data to test its predictions.

My job market paper and main research agenda study how people learn directly from data, how they form representations of how the world works, and how features of the decision problem shape what they learn. In particular, I ask when people recover stable relationships that generalize across environments, and when they instead learn correlations specific to the data at hand. My job market paper, *Extracting Models from Data: A Cost-Benefit Framework* (with Patrick Sewell), develops a theoretical framework in which agents select which variables to use and extract behavioral rules directly from data. Rule extraction reflects a trade-off between the value of a rule and its cognitive cost, so that the values and complexity costs of competing rules jointly shape what agents learn. By linking the primitives of the decision problem to testable predictions for rule extraction, the framework yields comparative statics showing how incentives, the number of available variables, and the statistical properties of the dataset interact with complexity costs to shape learning.

In a controlled experiment, we find strong support for these predictions: people are substantially more likely to recover the highest-value rule when its advantage over competing rules is larger, even when its own predictive accuracy remains approximately unchanged. Because rule-specific costs can be identified from observed extraction probabilities and rule values, we can recover these costs from choice data and distinguish which features of a learning problem enter the cost function from those that instead operate by changing the relative values of competing rules. Finally, through the lens of the framework, we show that heterogeneity in rule extraction across agents with different cognitive costs is systematic and predictable: lower-cost agents recover the highest-value rule more often and, when they make mistakes, place more weight on higher-value alternatives, generating systematic disagreement even among individuals facing the same data.

I plan to extend this agenda in three directions. First, in a direct extension of the JMP, I relax the assumption that the data from which individuals learn are exogenous. In a new preliminary project, *"Learning Rules from Endogenous Data"*, I study settings in which past choices shape the data used to evaluate behavioral rules. I characterize when this feedback generates learning traps, with a key distinction arising from what individuals observe. When they see only the resulting data, even objectively correct evaluation can become self-confirming, allowing welfare-inferior rules to persist. When they also observe which rule generated the data, they can evaluate each rule in the environment it creates rather than only on the inherited mix of environments. Preserving this provenance eliminates the self-confirming performance feedback and can restore a unique long-run outcome.

Second, building on the job market paper's broader agenda of deriving behavioral biases endogenously from features of the learning environment, in *"The Fundamental Attribution Error in Hiring Decisions"*

(with Eva Spantidaki Kyriazi), we study the tendency to attribute performance too strongly to worker ability and too little to task difficulty—the fundamental attribution error (FAE). We develop a theoretical framework that goes beyond documenting the FAE or taking it as a primitive of the model, and instead derives it from the statistical properties of the data and agents' cognitive abilities, rather than treating it as a stable psychological disposition. Specifically, we show that the emergence and magnitude of the FAE are shaped by the relative predictive power of worker ability and task difficulty in the data, while heterogeneity in information-processing costs generates systematic differences in the extent of the bias.

Finally, in line with the importance of learning to generalize emphasized in my job market paper, a final project, *"Learning to Generalize: An Information-Theoretic Approach"*, studies how the structure of variation across training datasets shapes generalization. Learners may memorize patterns that work in the environments they have seen, or instead extract a rule that remains valid when superficial features change. I develop an information-theoretic framework showing when variation across training environments promotes this second form of learning. When the same underlying structure appears across different settings, memorizing each setting separately requires maintaining an increasing number of distinctions, while representing the common rule becomes relatively less costly. This logic yields quantitative measures of variability in the learning sample, based on *logical entropy*, that identify which training experiences are most conducive to generalization to new settings.

Whereas my work on learning asks how the primitives of the environment shape what individuals learn, **a second research agenda studies how context shapes choice**. In *"Choice by Contextual Associations"* (with Jose Apesteguia), we develop a framework in which features of the environment activate associations that change the value of alternatives. We represent a context by the topics active in the decision environment, such as fairness, patriotism, or democracy in a political campaign, and model associations as links between those topics and the available alternatives. An alternative's value then combines a context-independent intrinsic component with the associations triggered by those topics: a voter may have an intrinsic preference for a party, while the party's appeal is further shaped by its associations with patriotism, fairness, or democracy. This formulation turns an intuitive psychological mechanism into a disciplined choice-theoretic model: we characterize its empirical content, establish conditions for identification, and derive comparative statics. We also show how the model's contextual primitives can be recovered from unstructured data, using large language models to identify active topics in text and images. This makes the relationship between context and choice empirically measurable.

This paper reflects the kind of formal work I most enjoy: starting from a broad behavioral regularity, identifying a clear psychological and economic mechanism, and deriving its testable implications from first principles. The next project extends this agenda by decomposing the effect of context on choice into two economically distinct channels: changes in preferences and changes in beliefs. With Jose Apesteguia, we plan to develop a falsifiable framework that separates contextual associations operating through beliefs from those operating through utility. This distinction matters for identifying how narratives operate, since narratives may shift both beliefs and utility, and for information policy, since information interventions may change beliefs without necessarily changing utility. We also plan to

test this distinction experimentally by manipulating context while separately measuring beliefs and valuations, allowing us to identify which channel drives the resulting changes in choice.

The role of beliefs provides a natural bridge to a third stream of my research agenda, which **studies cooperation and repeated games, with a focus on how beliefs about others' behavior and the resulting strategic uncertainty shape choices**. In *"The Importance of Being Even: Restitution and Cooperation"* (with Maria Bigoni, Marco Casari, Andrzej Skrzypacz, and Giancarlo Spagnolo; revise and resubmit at *AEJ: Microeconomics*), we study how cooperation can be restored in repeated interactions after a deliberate or accidental breach. We introduce restitution strategies, which repair the imbalance created by a deviation before returning to cooperation, and theoretically characterize a particular strategy that we call Payback. Combining formal equilibrium analysis with a large-scale reanalysis of repeated-game experiments, we find broad support for restitution strategies and, through structural estimation, for Payback in particular.

In *"Strategic Risk in Repeated Games with Imperfect Monitoring"*, I study repeated games with imperfect monitoring, in which noisy signals create a fundamental trade-off: because bad signals arise on the cooperative equilibrium path, sustaining valuable interaction requires lenient and forgiving strategies, but these strategies also leave players more exposed to opportunistic behavior. I formalize this trade-off and use existing experimental data to study the strategies individuals adopt. I find that individuals do not account for strategic risk and, conditional on cooperating, choose strategies that maximize equilibrium value.

Finally, in *"Self-Projection and Sorting Across Environments"*, I ask whether giving people a choice between high-surplus interactions vulnerable to free riding and safer, lower-return alternatives can sustain cooperation even without repeated interaction or formal punishment. The answer is not obvious: cooperators may avoid the high-surplus option to protect themselves from exploitation, while non-cooperators may select into it precisely to exploit others' cooperation. I show theoretically that taste projection, the tendency to expect others to share one's own cooperative preferences, resolves this ambiguity in favor of cooperation. Cooperative types expect more cooperation and select into the high-surplus environment, while non-cooperative types expect widespread defection and increasingly choose the safer alternative. Pilot data from an online experiment provide preliminary evidence for both the effect on cooperation and the proposed taste-projection mechanism.

I plan to bring these projects to completion and submission in the near term, while continuing to develop a broader research agenda on how people learn from their environments and how those environments shape choice.